Einstein/CTensorNP

Setup

In[ ]:=
```
<< mGRG`Einstein`
```

In[ ]:=
```
$PreRead = ReplaceAll[#,
    expr_String :> StringReplace[expr, "EP" -> "mGRG`Einstein`Private"]] &;
```

BasisNP: Calculate Newman-Penrose Frame

In[ ]:=
```
BasisNP::usage
```

Out[ ]=
```
BasisNP[metric, simpCmd] generates a Newman-Penrose
  null tetrad {l, n, m, m-bar} from a 4x4 metric matrix.
  'simpCmd' is an optional simplification function.
```

Metric을 인자로 하여 Newman-Penrose 기준 벡터를 얻는다. 옵션으로 simplification을 위한 함수가 될 수 있다. (See Chandrasekhar, pp 134)

In[ ]:=
```
coSys = {t, r, θ, ϕ};
metric =
  {{-1 + 2 M/r, 0, 0, 0}, {0, 1/(1 - 2 M/r), 0, 0}, {0, 0, r^2, 0}, {0, 0, 0, r^2 Sin[θ]^2}};
```

In[ ]:=
```
nullVs = BasisNP[metric, Simplify]; MatrixForm[nullVs]
```

Out[ ]//MatrixForm=

$$\begin{pmatrix} \frac{r}{2\,M-r} & 1 & 0 & 0 \\ -\frac{1}{2} & -\frac{1}{2}+\frac{M}{r} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{\frac{1}{r^2}}}{\sqrt{2}} & \frac{\sqrt{-\frac{\text{Csc}[\theta]^2}{r^4}}}{\sqrt{2}\,\sqrt{\frac{1}{r^2}}} \\ 0 & 0 & \frac{\sqrt{\frac{1}{r^2}}}{\sqrt{2}} & -\frac{\sqrt{-\frac{\text{Csc}[\theta]^2}{r^4}}}{\sqrt{2}\,\sqrt{\frac{1}{r^2}}} \end{pmatrix}$$

BasisNP의 결과값을 확인한다.

In[ ]:=
```
MatrixForm[Table[nullVs[[i]] . metric . nullVs[[j]], {i, 4}, {j, 4}] // Simplify]
```

Out[ ]//MatrixForm=

$$\begin{pmatrix} 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

In[ ]:=
```
nullVs = PowerExpand[nullVs]
```

Out[ ]=

$$\left\{\left\{\frac{r}{2\,M-r},\,1,\,0,\,0\right\},\,\left\{-\frac{1}{2},\,-\frac{1}{2}+\frac{M}{r},\,0,\,0\right\},\,\left\{0,\,0,\,\frac{1}{\sqrt{2}\,r},\,\frac{i\,\text{Csc}[\theta]}{\sqrt{2}\,r}\right\},\,\left\{0,\,0,\,\frac{1}{\sqrt{2}\,r},\,-\frac{i\,\text{Csc}[\theta]}{\sqrt{2}\,r}\right\}\right\}$$

RotateNP: Basis Transformation

In[ ]:=
```
RotateNP::usage
```

Out[ ]=

```
RotateNP[nullVectors, classN, p1, p2, simpCmd] performs
  a Lorentz rotation on the given 'nullVectors'. 'classN'
  specifies the rotation type (1: null rotation about l, 2: null
  rotation about n, 3: spin-boost). 'p1' and 'p2' are rotation
  parameters. 'simpCmd' is an optional simplification function.
```

Newman-Penrose 기준 벡터를 변환한다. 첫 번째 인자는 기준 벡터이고, 두 번째 인자는 변환 Class이다. 세 번째와 네 번째 인자는 변환 변수이다. 옵션으로 simplification을 위한 함수가 될 수 있다.

In[ ]:=
```
RotateNP[nullVs, 3, r/(r - 2 M), 0, Simplify]
```

Out[ ]=

$$\left\{\left\{-1,\,1-\frac{2\,M}{r},\,0,\,0\right\},\,\left\{\frac{r}{4\,M-2\,r},\,-\frac{1}{2},\,0,\,0\right\},\,\left\{0,\,0,\,\frac{1}{\sqrt{2}\,r},\,\frac{i\,\text{Csc}[\theta]}{\sqrt{2}\,r}\right\},\,\left\{0,\,0,\,\frac{1}{\sqrt{2}\,r},\,-\frac{i\,\text{Csc}[\theta]}{\sqrt{2}\,r}\right\}\right\}$$

Initialization of Component Tensors in NP Formalism

In[ ]:=
```
InitCTensorNP::usage
```

Out[ ]=

```
InitCTensorNP[coSys, nullVectors, opts] initializes
  the environment for Newman-Penrose formalism calculations.
  'coSys' is a list of 4 coordinates, and 'nullVectors' is
  the 4x4 null tetrad matrix {l, n, m, m-bar}. Options can
  be used to control re-initialization and simplification.
```

좌표계와 기준 벡터를 입력으로 받아 Newman-Penrose 방식으로 성분 텐서를 초기화 한다. 이 함수는 SpinCoefficients와 곡률 텐서를 자동적으로 계산하고, CoordinateBasisFlag과 KdeltaFlag을 Off 시킨다. 옵션으로 SimplifyMore가 있다.

In[ ]:=
```
coSys = {t, r, θ, ϕ};
nullVs = {{r/(2 M - r), 1, 0, 0}, {-1/2, -1/2 + M/r, 0, 0},
    {0, 0, 1/(Sqrt[2] r), I Csc[θ]/(Sqrt[2] r)}, {0, 0, 1/(Sqrt[2] r), -I Csc[θ]/(Sqrt[2] r)}};
```

In[ ]:=
```
Timing[InitCTensorNP[coSys, nullVs, SimplifyMore → True]]
```

Out[ ]=
```
{0.052576, Null}
```

SpinCoefficients, PhiNP, PsiNP, PetrovType의 값을 보여 준다.

In[ ]:=
```
Show[SpinCoefficients]
```

$$\epsilon = \kappa = \lambda = \nu = \pi = \sigma = \tau = 0$$

$$\alpha = \frac{\text{Cot}[\theta]}{2\sqrt{2}\,r}$$

$$\beta = -\frac{\text{Cot}[\theta]}{2\sqrt{2}\,r}$$

$$\gamma = -\frac{M}{2r^2}$$

$$\mu = \frac{-2M + r}{2r^2}$$

$$\rho = \frac{1}{r}$$

In[ ]:=
```
Show[PhiNP]
```

$$\Phi_{00} = \Phi_{01} = \Phi_{02} = \Phi_{11} = \Phi_{12} = \Phi_{22} = \Lambda = 0$$

In[ ]:=
```
Show[PsiNP]
```

$$\Psi_0 = \Psi_1 = \Psi_3 = \Psi_4 = 0$$

$$\Psi_2 = -\frac{M}{r^3}$$

In[ ]:=
```
Show[PetrovType]
```

```
Petrov Type: D
```

In[ ]:=
```
{WeylCD[-1, -2, -1, -2],
 RiemannCD[-1, -2, -1, -2], RicciCD[-1, -1], ScalarCD[]}
```

Out[ ]=
```
{-2 M/r^3, -2 M/r^3, 0, 0}
```

Clear Component Tensors in NP Formalism

In[ ]:=
```
ClearCTensorNP[]
```

In[•]:=
```
{WeylCD[-1, -2, -1, -2], PsiNP[0], PhiNP[0, 1], LambdaNP}
```

Out[•]=
$$\{C_{1212}, \text{PsiNP}[0], \text{PhiNP}[0, 1], \text{LambdaNP}\}$$

In[•]:=
```
{RicciCD[-1, -1], ScalarCD[], RiemannCD[-1, -2, -1, -2]}
```

Out[•]=
$$\{R_{11}, R, R_{1212}\}$$

In[•]:=
```
{EP`epsilonNP, EP`kappaNP, EP`piNP, EP`gammaNP, EP`tauNP, EP`nuNP}
```

Out[•]=
```
{mGRG`Einstein`Private`epsilonNP, mGRG`Einstein`Private`kappaNP,
 mGRG`Einstein`Private`piNP, mGRG`Einstein`Private`gammaNP,
 mGRG`Einstein`Private`tauNP, mGRG`Einstein`Private`nuNP}
```

In[•]:=
```
{EP`betaNP, EP`sigmaNP, EP`lambdaNP, EP`rhoNP, EP`muNP, EP`alphaNP}
```

Out[•]=
```
{mGRG`Einstein`Private`betaNP, mGRG`Einstein`Private`sigmaNP,
 mGRG`Einstein`Private`lambdaNP, mGRG`Einstein`Private`rhoNP,
 mGRG`Einstein`Private`muNP, mGRG`Einstein`Private`alphaNP}
```

In[•]:=
```
{GammaCD[-1, -1, -1], GammaCD[-1, -1, 1]}
```

Out[•]=
$$\{\Gamma_{111}, \Gamma_{11}{}^{1}\}$$

In[•]:=
```
{Structuref[-1, -2, -1], Structuref[-1, -2, 1]}
```

Out[•]=
$$\{f_{121}, f_{12}{}^{1}\}$$

In[•]:=
```
{EP`inverseBasisMatrix, mGRG`STensor`Private`basisMatrix[DefaultKind]}
```

Out[•]=
```
{mGRG`Einstein`Private`inverseBasisMatrix,
 mGRG`STensor`Private`basisMatrix[Latin]}
```

In[•]:=
```
{Metricg[-1, -1], Metricg[1, 1]}
```

Out[•]=
$$\{g_{11}, g^{11}\}$$

In[•]:=
```
{EP`nDimension, GetDimension[], GetCoordinates[DefaultKind]}
```

Out[•]=
```
{mGRG`Einstein`Private`nDimension,
 GetDimension[Latin], GetCoordinates[Latin]}
```

In[ ]:=
```
{mGRG`STensor`Private`flagTable[InitCTensorFlag],
 mGRG`STensor`Private`flagTable[EvaluateBDFlag]}
```

Out[ ]=
```
{False, False}
```

In[ ]:=
```
{EP`defaultCTensor[EP`simpMethod]}
```

Out[ ]=
```
{mGRG`Einstein`Private`defaultCTensor[mGRG`Einstein`Private`simpMethod]}
```